User Manual

This is the Drug Development Portfolio Dashboard, a management system designed to track clinical programs through various stages of the R&D (Research and Development) lifecycle.

1. Core Navigation & High-Level Metrics

The dashboard is divided into several functional modules accessible via the top navigation bar:

- Programs: The main view (currently active), listing specific drug candidates.
- Studies: Likely focuses on individual clinical trials within those programs.
- Analytics & Compliance: For tracking success rates and regulatory adherence.
- Budget: For monitoring financial allocation across the portfolio.

2. Portfolio Filtering

On the left sidebar, you can drill down into the data using two primary categories:

- Development Phase: Tracks the pipeline from early Discovery and Preclinical work through Phase 1-3 trials, up to NDA Review and final Approved status.
- Therapeutic Area: Segregates programs by medical specialty, such as Oncology, Neurology, or Cardiology.

3. Program Table (The "Workhorse")

The main table provides a real-time snapshot of 50 active programs. Each entry includes:

- Program Name & ID: The internal name and manufacturer code (e.g., *Solmix GSK-6578*).
- Current Phase: Where the drug sits in the regulatory timeline.
- Milestone: The specific goal currently being pursued (e.g., *Compound Synthesis*, *Efficacy Study*, or *Regulatory Submission*).
- Status: A color-coded health indicator:
    - Active/Completed: Proceeding as planned.
    - On-Hold: Temporarily paused.
    - Discontinued: Removed from the pipeline (often due to safety or efficacy issues).

4. Summary of Pipeline Health

Looking at the data, the portfolio appears diverse but carries a high volume of discontinued and on-hold projects in the current view. This is typical in drug development, where high "attrition rates" are expected as drugs move from the lab into human testing.

Architecture

Frontend (Next.js + React)

- Pages & Routing (app): Multiple feature sections:
  - Programs, Studies, Analytics, Budget, Compliance
  - Admin panel for management
- Redux State Management (store): Manages programs, filters, authentication, theme, analytics, search
- Shared Components (shared): Reusable UI components (Navigation, Login, Layout, etc.)
- Feature Modules (features): Organized by domain with their own components and services
  - Programs (filtering, listing, details)
  - Analytics, Budget, Compliance, Studies, Admin

Backend (API Routes)

- api: Next.js API routes for data
  - /programs - Get all/create programs
  - /programs/[id] - Get/update specific program
  - /analytics, /studies - Domain-specific endpoints
- server: Services and repositories for business logic and data access

Key Features

- Dark/Light Theme toggle with CSS variables
- Advanced Filtering (development phase, therapeutic area)
- Drill-down Details for individual programs and studies
- Metadata Editing for authorized users
- Responsive Design (desktop, tablet, mobile)
- Modern UI with gradients, shadows, smooth transitions
- Accessibility (WCAG-compliant, semantic HTML, keyboard navigation)

Tech Stack

- Next.js 15 (App Router)
- TypeScript
- Redux Toolkit for state management
- CSS Modules with CSS variables for theming
- React 18